

The exceptional ℓ_∞ -sum: a convex-DLD2P counterexample and the exact inheritance characterization

Candidate result from run `fa_banach_001`, lane 10

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Status. Candidate counterexample and full characterization, likely valid, pending human expert review. The proof is non-computational.

1 Source problem

The source is T. A. Abrahamsen, R. Haller, V. Lima, and K. Pirk, *Delta- and Daugavet-points in Banach spaces*, arXiv:1812.02450 [1]. Problem 1 on printed page 21 asks:

Given $X \oplus_N Y$ with a Δ -point/a Daugavet point/the convex DLD2P, does X have a Δ -point/a Daugavet point/the convex DLD2P?

This packet settles the convex-DLD2P branch negatively and, together with a recent positive result, gives the exact characterization of the absolute norms for which factor inheritance always holds.

Remark 5.9. Let X and Y be Banach spaces. If X has the convex DLD2P and N is the ℓ_∞ -norm, then $X \oplus_N Y$ has the convex DLD2P.

Although we have mostly settled the results about the question whether the direct sum with absolute normalized norm has a Δ -point/a Daugavet-point/the convex DLD2P (there are some norms left to look at in the Daugavet-point case), the results about the components of a direct sum with a given property having the same property, are all still unknown.

PROBLEM 1. *Given $X \oplus_N Y$ with a Δ -point/a Daugavet point/the convex DLD2P, does X have a Δ -point/a Daugavet point/the convex DLD2P?*

Figure 1: Problem 1 and its immediate context in [1, p. 21].

2 Definitions and input results

All spaces are real. For a Banach space W , write B_W and S_W for its closed unit ball and unit sphere, and write $\Delta_W \subseteq S_W$ for its set of Δ -points. Thus $w \in S_W$ is a Δ -point precisely when, for every slice S of B_W containing w and every $\varepsilon > 0$, there is $u \in S$ such that $\|w - u\| \geq 2 - \varepsilon$. The space W has the convex-DLD2P when

$$B_W = \overline{\text{co}}(\Delta_W).$$

For $k > 1$, Haller, Pirk, and Veeorg call $w \in S_W$ a Δ_k -point if every inclusion $w \in S(B_W, w^*, \alpha)$ and every $\varepsilon > 0$ admit

$$u \in S(B_W, w^*, k\alpha), \quad \|w - u\| \geq 2 - \varepsilon.$$

Every Δ -point is a Δ_k -point for every $k > 1$.

We use the following three results from [2, Theorem 4.1, Example 4.3, Proposition 4.4].

1. If (ax, by) is a Δ -point of $X \oplus_N Y$, where $a, b \geq 0$, $N(a, b) = 1$, $x \in S_X$, $y \in S_Y$, and $b \neq 1$, then x is a Δ -point of X .
2. If $x \in S_X$ is not a Δ -point and $y \in S_Y$ is a Δ -point, then for every $k > 1$

$$((1 - k^{-1})x, k^{-1}y) \quad \text{is a } \Delta_k\text{-point of } X \oplus_1 Y.$$

3. If $p, q > 1$, $p^{-1} + q^{-1} = 1$, x is a Δ_p -point of X , and y is a Δ_q -point of Y , then (x, y) is a Δ -point of $X \oplus_\infty Y$.

Finally, $C[0, 1]$ has the Daugavet property, so every point of its unit sphere is a Daugavet point and hence a Δ -point; see, for example, [4, Section 2].

3 Idea of the proof

Let $D = C[0, 1]$ and first form $E = \mathbb{R} \oplus_1 D$. The scalar factor has no Δ -points. Consequently the only genuine Δ -points of E lie on the D -axis, so E fails the convex-DLD2P. Nevertheless, Example 4.3 of [2] supplies many weaker Δ_k -points in E : they approach both ends of the missing scalar axis as $k \rightarrow \infty$. Together with the genuine Δ -points on the D -axis, these Δ_k -points convexly generate all of B_E .

Now take $Z = E \oplus_\infty D$. Every unit vector of the second copy of D is a Δ_q -point for every $q > 1$. Pairing it with any one of the above Δ_k -points of E , with conjugate exponents, produces a genuine Δ -point of Z by Proposition 4.4 of [2]. A product-of-convex-hulls identity then shows that these points convexly generate the entire ball $B_Z = B_E \times B_D$.

4 Counterexample and characterization

Theorem 1. *Let*

$$D = C([0, 1]; \mathbb{R}), \quad E = \mathbb{R} \oplus_1 D, \quad Z = E \oplus_\infty D.$$

Then

1. $\Delta_E = \{0\} \times S_D$, and consequently E does not have the convex-DLD2P;
2. Z has the convex-DLD2P.

Thus the convex-DLD2P of an ℓ_∞ -sum need not pass to its factors.

Moreover, for an absolute normalized norm N on $\mathbb{R}^2$, the following are equivalent:

- (a) *for all real Banach spaces X, Y , convex-DLD2P of $X \oplus_N Y$ implies convex-DLD2P of both X and Y ;*
- (b) *N is not the ℓ_∞ -norm.*

Proof. Every $d \in S_D$ is a Δ -point. The standard component-to-sum slice argument therefore shows that $(0, d)$ is a Δ -point of $E = \mathbb{R} \oplus_1 D$.

Conversely, suppose $(r, d) \in S_E$ is a Δ -point and $r \neq 0$. Put $a = |r|$ and $b = \|d\|$, so $a + b = 1$ and $b < 1$. Writing $r = a \operatorname{sgn}(r)$ (and, when $d \neq 0$, $d = b(d/\|d\|)$), input result 1 implies that $\operatorname{sgn}(r)$ is a Δ -point of $\mathbb{R}$. This is impossible: a sufficiently small slice of $B_{\mathbb{R}} = [-1, 1]$ containing 1 (respectively -1) contains no point almost at distance two from it. The same invocation allows an arbitrary unit vector in the zero-coordinate when $d = 0$. Hence

$$\Delta_E = \{0\} \times S_D.$$

In particular, the continuous functional $(r, d) \mapsto r$ vanishes on $\overline{\operatorname{co}}(\Delta_E)$ but not on $(1, 0) \in B_E$, so E fails the convex-DLD2P.

For $k > 1$, $\sigma \in \{-1, 1\}$, and $d \in S_D$, set

$$e_{k,\sigma,d} := ((1 - k^{-1})\sigma, k^{-1}d) \in S_E$$

and define

$$A := \Delta_E \cup \{e_{k,\sigma,d} : k > 1, \sigma \in \{-1, 1\}, d \in S_D\}.$$

By input result 2, $e_{k,\sigma,d}$ is a Δ_k -point of E . Every member of Δ_E is a Δ_p -point for every $p > 1$. Thus every $e \in A$ is a $\Delta_{p(e)}$ -point for at least one $p(e) > 1$.

Fix $d_0 \in S_D$. As $k \rightarrow \infty$,

$$e_{k,\sigma,d_0} \longrightarrow (\sigma, 0),$$

so both ends of the scalar axis belong to $\overline{A}$. The whole D -axis $\{0\} \times S_D$ already lies in A . Since the unit ball of an ℓ_1 -sum is the closed convex hull of the unit balls on its two coordinate axes, it follows that

$$B_E = \overline{\operatorname{co}}(A). \tag{1}$$

Let $e \in A$ and $v \in S_D$. Choose $p = p(e) > 1$ such that e is a Δ_p -point, and let $q = p/(p-1)$. Since v is a genuine Δ -point, it is a Δ_q -point. Input result 3 gives

$$(e, v) \in \Delta_Z.$$

Therefore

$$A \times S_D \subseteq \Delta_Z. \tag{2}$$

For any nonempty subsets $A_0 \subseteq B_{W_0}$ and $A_1 \subseteq B_{W_1}$ one has

$$\overline{\operatorname{co}}(A_0 \times A_1) = \overline{\operatorname{co}}(A_0) \times \overline{\operatorname{co}}(A_1).$$

Indeed, one inclusion is immediate, while finite convex combinations $x = \sum_i \alpha_i a_i$ and $y = \sum_j \beta_j b_j$ satisfy

$$(x, y) = \sum_{i,j} \alpha_i \beta_j (a_i, b_j),$$

and passage to closures gives the reverse inclusion. Using (1), $B_D = \overline{\operatorname{co}}(S_D)$, and (2), we obtain

$$B_Z = B_E \times B_D = \overline{\operatorname{co}}(A \times S_D) \subseteq \overline{\operatorname{co}}(\Delta_Z) \subseteq B_Z.$$

Thus Z has the convex-DLD2P, proving the counterexample.

For the final equivalence, Guerrero-Viu and Markowicz prove in [3, Proposition 2] that whenever $N \neq \|\cdot\|_\infty$, convex-DLD2P passes from $X \oplus_N Y$ to both factors. This is (b) $\Rightarrow$ (a). The example above shows that (a) fails for $N = \|\cdot\|_\infty$, proving the converse. $\square$

5 Verification notes

- The two possible hidden boundary cases in the description of Δ_E were checked: $d = 0$ is covered by Theorem 4.1 of [2], and $r = 0$ gives genuine Δ -points by the slice characterization.
- Every point used in A lies on S_E . The limits $e_{k,\sigma,d_0} \rightarrow (\sigma, 0)$ recover precisely the missing scalar endpoints.
- The conjugate exponent attached to $e_{k,\sigma,d}$ is $q = k/(k-1) > 1$. A genuine Δ -point is a Δ_q -point for this (possibly k -dependent) value of q .
- The product convex-hull equality is justified by the double weights $\alpha_i\beta_j$; no synchronized decompositions are assumed.
- No numerical or computer-assisted step is used.

6 Novelty search and limitations

A bounded search was performed on 9 August 2026 in the run’s registry, solution, attempt, and proof-gap indexes; arXiv/source text for 1812.02450, 2001.06197, and 2607.01875; and web/arXiv queries combining “convex-DLD2P”, “ ℓ_∞ ”, “absolute sum”, “factor”, “counterexample”, and “ Δ_k ”. No prior statement of this counterexample or of the resulting if-and-only-if characterization was found. Most significantly, the recent paper [3], submitted 2 July 2026, proves the positive result only for $N \neq \ell_\infty$ and explicitly identifies the ℓ_∞ -sum as the exceptional unresolved case.

This is nevertheless a bounded novelty check, not a claim of exhaustive priority. The packet settles only the convex-DLD2P branch of source Problem 1; its questions about existence of a Δ -point or a Daugavet point are separate. Human review should focus on the use of the three Δ_k results from [2] and the closure identity in the last step.

References

- [1] T. A. Abrahamsen, R. Haller, V. Lima, and K. Pirk, *Delta- and Daugavet-points in Banach spaces*, Proc. Edinburgh Math. Soc. **63** (2020), 475–496; arXiv:1812.02450.
- [2] R. Haller, K. Pirk, and T. Veeorg, *Daugavet- and Delta-points in absolute sums of Banach spaces*, J. Convex Anal. **28** (2021), no. 3, 909–924; arXiv:2001.06197.
- [3] J. Guerrero-Viu and J. Markowicz, *On super Delta-points and the convex-DLD2P in absolute sums*, arXiv:2607.01875v1 (2026).
- [4] V. M. Kadets, M. I. Shvidkoy, G. G. Sirotkin, and D. Werner, *Banach spaces with the Daugavet property*, Trans. Amer. Math. Soc. **352** (2000), 855–873.